

The Shape of a Knowledge Boundary: A Transformer Trained Only on Pre-1939 Text

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Code (GitHub): <https://github.com/Shinoaki798/1939> · I used Claude Fable 5.1 to do the proposal.

1 Introduction

A language model’s knowledge cutoff is usually described as a wall: everything before, nothing after. I suspect it is a **membrane with structure**: some post-cutoff concepts are unreachable, others are assembled from parts the model already has and are half predictable. To characterise that structure I will train **one Transformer network**, from random initialisation, on newspaper text ending strictly before the outbreak of the Second World War — enlarged, as a second corpus stage, with pre-cutoff German and French press machine-translated into English — and use that single model to answer three questions. **RQ1 (permeability)**: which classes of post-1939 concepts can a pre-1939 model reach? **RQ2 (invertibility)**: can the boundary be inverted into a tool that dates a passage and audits a corpus for post-1939 text? **RQ3 (foresight by perspective)**: how much of 1939–45 was already implied by the press of mid-1939 — American, German or French? Deep learning is the instrument: these quantities exist only in a trained model’s next-token distribution, so the project is the training, validation and testing of that model. *Boundary*: training ends **30 June 1939**; 1 July–31 August 1939 is an **embargo** excluded from training and evaluation, so the August crisis cannot leak in.

2 Illustration

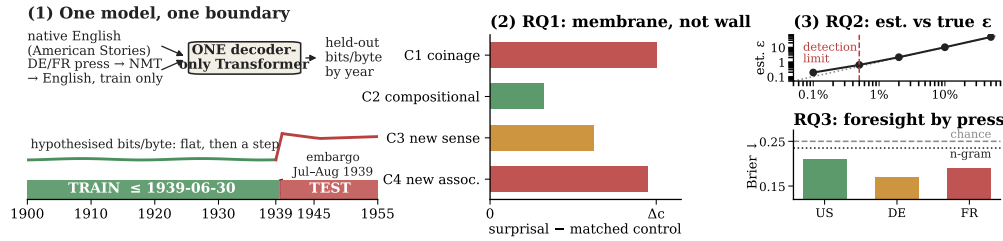


Figure 1: One model, three measurements. Shapes are hypothesised, not results.

3 Background and Related Work

Language models degrade on text written after their training window [7]; temporal conditioning improves calibration on future facts [2]; Time Machine GPT [3] trains point-in-time models uninformed of later facts. Several recent projects train models from scratch on historically bounded corpora at cutoffs across the 19th and early 20th centuries, including 1939 [4, 5, 11]; they aim to *produce* such models; this project studies the boundary itself. SemEval-2020 Task 1 [12], HistBERT [10] and nineteenth-century models [6] measure how *existing* words drift, using masked encoders; RQ1 asks which *not-yet-existing* concepts are reachable from a fixed earlier vocabulary, which needs a causal model with a hard boundary. ChroniclingAmericaQA [9] shows a large accuracy drop on uncorrected OCR, motivating OCR quality as a covariate.

4 Data Processing

Sources, in two stages. *Corpus v1* (frozen 10 Oct): American Stories [1], article-level (not page-OCR) text from Chronicling America. *Corpus v2* (frozen 24 Oct) adds non-English pre-cutoff press translated to English: (i) foreign-language titles inside Chronicling America (same metadata and OCR); (ii) Gallica (FR) and the Deutsches Zeitungsportal (DE) for the domestic press RQ3 needs — its ~ 180 context articles are translated first, independently of v2; (iii) Chinese press, as a stretch. The Transformer trains on v2 if frozen in time, else on v1 with translation reported as an extension, so no graded result depends on it. **Splits** are by publication date, never by shuffling: from every year

1900–1955 a fixed 2% of articles is held out first and never trained on (the per-year curve); TRAIN = the rest $\leq 1939-06-30$; VAL = a disjoint 2% $\leq 1939-06-30$; TEST-A/B/C = held-out 1930–38 / 1939-09–1945 / 1946–55. MinHash dedup of wire-service reprints runs *before* splitting. **Cleaning.** Per-article OCR quality (period-lexicon hit rate) is a covariate reported by year and language, with a fixed drop threshold. **Translation** is data scale, not cleaning: unique pre-1939 tokens bound how GPT-2-like the model can get, and translation is the only lever that enlarges them. A fixed-version sentence-level NMT model (NLLB [8]; not an LLM, which could add hindsight) runs on a second machine in parallel, with three defences against its 2020s English: (1) any translated sentence containing an RQ1 probe term is dropped whole; (2) translated text is never *scored*: it serves as training data and as RQ3 conditioning context only, and every scored evaluation set is native English; (3) its residual probe hit rate is reported, and a held-out translated slice is scored against native text (the translation-dialect effect). No cap on the translated share; throughput is the bound. **Tokenizer:** 32k BPE on native-English pre-cutoff text only; all numbers are bits-per-byte. **RQ1 probe set:** ~ 200 post-cutoff terms in four classes — C1 coinage (*radar, genocide*), C2 compositional (*atomic bomb*), C3 old word, new sense (*blitz, resistance*), C4 existing proper noun, new association (*Pearl Harbor, Hiroshima*) — each with a frequency-matched pre-cutoff control in ≥ 5 sentence contexts, frozen in advance. **RQ2 calibration:** held-out pre-cutoff corpora with a known fraction ε (0.1–50%) of 1939-09–1945 articles mixed in. **RQ3 propositions:** ~ 60 binary propositions about 1939–45, each with three same-topic contexts (US, DE, FR press) from June 1939 and from late August 1939 — the embargo, used as conditioning text only, by explicit exception.

5 Architecture

One **decoder-only Transformer network** as in the Transformer lecture: token embeddings, stacked blocks of multi-head self-attention and position-wise feed-forward layers, residual connections, layer normalisation, softmax over the vocabulary, with attention **causally masked** (position i sees only $\leq i$) so it is a next-token predictor. Decoder-only is dictated by the measurements, all sequence probabilities of unseen text: a masked encoder’s pseudo-likelihoods neither form a sequence probability nor score a continuation, and an encoder–decoder has no source sequence here. This is the GPT-2 configuration. No pretrained weights, and **no second Transformer** as control or ablation: all controls are built at evaluation time on this one model. From scratch in PyTorch, context 1024. Size follows data at ~ 20 tokens/parameter, **never over two epochs** (repetition raises memorisation, biasing surprisal): 90M/160M/250M/335M parameters at $< 1.5\text{B} / 1.5\text{--}3\text{B} / 3\text{--}6\text{B} / > 6\text{B}$ tokens; the largest (24 layers, $d=1024$, 16 heads) is $\approx 1.4 \times 10^{19}$ FLOPs, 30–40 h on one RTX 5080, finished before the midterm. **Measurement.** For term class c , $\Delta_c = \mathbb{E}_{t \in \mathcal{T}_c}[s(t)] - \mathbb{E}_{t \in \mathcal{C}_c}[s(t)]$, s the model’s surprisal in fixed contexts, $\mathcal{C}_c$ the matched controls. RQ1 is the profile $(\Delta_{C1..C4})$; RQ2 fits a monotone map to ε and reports the minimum detectable ε ; RQ3 scores forced-choice continuations by log-likelihood ratio as Brier score per perspective; all with bootstrap 95% CIs. **Success:** Δ_{C2} significantly smallest; a step at 1939, not a slope; $\hat{\varepsilon}$ monotone; perspectives above the n -gram floor and significantly different.

6 Baseline Model

On the identical training split, tokenizer and token budget, following the course progression: a **vanilla RNN**, an **LSTM** and a **GRU** language model, each 2-layer and $\sim 50\text{M}$ parameters, same next-token objective; the Transformer’s choice as primary model is justified by these comparisons. A **modified Kneser–Ney 5-gram** (KenLM defaults) is the non-neural floor; it cannot compose unseen compounds, so it should show no C1/C2 ordering. All are retrained on the final corpus for reported comparisons. **Plan** (individual): baselines 17–24 Oct; the Transformer run 25 Oct–2 Nov; evaluation through November.

7 Ethical Considerations

Pre-1939 newspapers carry the racial, ethnic and gender language of their time; the model will reproduce it, which is documented, not sanitised. Translated 1930s German press includes state propaganda and NMT can misattribute words; corpus shares by source are reported and RQ3 results are framed as properties of a record, not verdicts on nations; the RQ2 tool could be misused as an accusation, so it ships with its detection limit.

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